



# SPACE DESIGN LOAD SUMMARY

2026-09-28

## Bathroom\_2

Level: Level 0 · Internal condition: TM59\_Bathroom

### DESIGN CRITERIA

|                            | Heating | Cooling |
|----------------------------|---------|---------|
| Room set point             | 16.0 °C | none    |
| Outdoor dry bulb           | —       | —       |
| Outdoor RH (coincident)    | —       | —       |
| <b>ROOM HUMIDITY</b>       |         |         |
| Humidification set point   | —       |         |
| lower RH limit             |         |         |
| Dehumidification set point | —       |         |
| upper RH limit             |         |         |

### SIZING (TAS DESIGN LOADS)

|                                                            | Heating  | Cooling  |
|------------------------------------------------------------|----------|----------|
| Design load                                                | 0 W      | 0 W      |
| Design load per area                                       | 0.0 W/m² | 0.0 W/m² |
| Sizing multiplier                                          | not set  | not set  |
| Design loads: from Tas sizing — date/currency not recorded |          |          |

### HEATING

|                     |                                 |                               |
|---------------------|---------------------------------|-------------------------------|
| Design day          | Leeds_TRY ANN HTG 100% CONDS DB |                               |
|                     | Design day                      | Full year                     |
| Peak sensible load  | 1,140 W                         | 104 W                         |
| Peak hour           | 23:00–24:00                     | 23 Dec 09:00–10:00 (HOY 8554) |
| Room dry bulb       | 16.0 °C                         | 16.0 °C                       |
| Room resultant      | 13.9 °C                         | 15.9 °C                       |
| Room RH             | 20 %                            | 35 %                          |
| Room humidity ratio | 2.2 g/kg                        | 3.9 g/kg                      |
| Outdoor dry bulb    | —                               | -2.3 °C                       |
| Outdoor RH          | —                               | 100 %                         |

### SENSIBLE LOAD COMPONENTS AT PEAK

| Component                    | Design day (W) | Full year (W) |
|------------------------------|----------------|---------------|
| Infiltration / ventilation   | -112           | -93           |
| Building heat transfer       | -1,023         | -3            |
| External conduction — opaque | -5             | -7            |

Zero at both peaks (not listed): Solar, Lighting, Occupancy (sensible), Equipment (sensible), Air movement (inter-zone), External conduction — glazing, Air handling unit. Latent gains: zero at both peaks (Occupancy, Equipment).  
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

### COOLING

|                                                                                                                                |                                     |           |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------|
| Design day                                                                                                                     | Leeds_TRY ANN CLG 0% CONDS DB=>GRad |           |
|                                                                                                                                | Design day                          | Full year |
| Peak sensible load                                                                                                             | 0 W                                 | 0 W       |
| 0 W: no cooling demand on the design day or over the full year, so no peak hour, room or outdoor state or component breakdown. |                                     |           |

### RESULTS

|                                                                                                                                                                                                                                                                                                                           | Heating            | Cooling            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|
| Status                                                                                                                                                                                                                                                                                                                    | Available          | Available          |
| Result source                                                                                                                                                                                                                                                                                                             | SAM.Analytical.Tas | SAM.Analytical.Tas |
| Read into model                                                                                                                                                                                                                                                                                                           | 27 Sep 2026 20:42  | 27 Sep 2026 20:42  |
| Matches current model                                                                                                                                                                                                                                                                                                     | not recorded       | not recorded       |
| Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year, and its hour of the year (HOY 1 = 1 Jan 00:00–01:00, HOY 8760 = 31 Dec 23:00–24:00).<br>Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied. |                    |                    |